

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 - 25 =$

(2) Factor: $y^2 + 6y + 8 =$

(3) Factor: $24x^2 + 6x =$

(4) Factor: $8y^2 + 16y =$

(5) Factor: $x^2 - 36 =$

(6) Factor: $x^2 - 16 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $12x^2 + 6x =$

(2) Factor: $y^2 - 16 =$

(3) Factor: $x^2 + 6x + 8 =$

(4) Factor: $x^2 + 6x + 5 =$

(5) Factor: $9y^2 + 9y =$

(6) Factor: $x^2 + 8x + 15 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 36 =$

(2) Factor: $12y^2 + 16y =$

(3) Factor: $x^2 + 7x + 10 =$

(4) Factor: $x^2 + 8x + 12 =$

(5) Factor: $y^2 + 2y + 1 =$

(6) Factor: $21x^2 + 18x =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 9 =$

(2) Factor: $6x^2 + 10x =$

(3) Factor: $y^2 - 81 =$

(4) Factor: $x^2 + 2x + 1 =$

(5) Factor: $y^2 - 81 =$

(6) Factor: $y^2 + 9y + 20 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 + 9x + 20 =$

(2) Factor: $x^2 + 10x + 24 =$

(3) Factor: $y^2 + 7y + 12 =$

(4) Factor: $8y^2 + 10y =$

(5) Factor: $x^2 + 6x + 8 =$

(6) Factor: $y^2 - 81 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 16 =$

(2) Factor: $y^2 - 25 =$

(3) Factor: $18y^2 + 15y =$

(4) Factor: $y^2 - 4 =$

(5) Factor: $12y^2 + 10y =$

(6) Factor: $6y^2 + 6y =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 - 36 =$

(2) Factor: $x^2 + 5x + 4 =$

(3) Factor: $21y^2 + 12y =$

(4) Factor: $x^2 - 64 =$

(5) Factor: $12x^2 + 18x =$

(6) Factor: $24y^2 + 20y =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $18y^2 + 9y =$

(2) Factor: $y^2 + 7y + 12 =$

(3) Factor: $12y^2 + 12y =$

(4) Factor: $x^2 - 81 =$

(5) Factor: $10y^2 + 6y =$

(6) Factor: $16x^2 + 20x =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 16 =$

(2) Factor: $16y^2 + 16y =$

(3) Factor: $x^2 - 25 =$

(4) Factor: $y^2 + 2y + 1 =$

(5) Factor: $21x^2 + 12x =$

(6) Factor: $y^2 + 7y + 10 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 - 25 =$

(2) Factor: $x^2 + 6x + 8 =$

(3) Factor: $y^2 + 8y + 12 =$

(4) Factor: $12x^2 + 4x =$

(5) Factor: $8y^2 + 10y =$

(6) Factor: $x^2 - 64 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 + 7y + 6 =$

(2) Factor: $24x^2 + 16x =$

(3) Factor: $y^2 - 49 =$

(4) Factor: $24y^2 + 16y =$

(5) Factor: $y^2 - 81 =$

(6) Factor: $12x^2 + 18x =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 + 8x + 15 =$

(2) Factor: $x^2 - 64 =$

(3) Factor: $y^2 + 8y + 15 =$

(4) Factor: $y^2 + 7y + 12 =$

(5) Factor: $12y^2 + 6y =$

(6) Factor: $y^2 + 6y + 5 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $32y^2 + 16y =$

(2) Factor: $12y^2 + 15y =$

(3) Factor: $y^2 + 4y + 3 =$

(4) Factor: $y^2 + 7y + 12 =$

(5) Factor: $8y^2 + 16y =$

(6) Factor: $20y^2 + 16y =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $21y^2 + 15y =$

(2) Factor: $x^2 + 6x + 9 =$

(3) Factor: $x^2 - 9 =$

(4) Factor: $x^2 + 3x + 2 =$

(5) Factor: $x^2 - 4 =$

(6) Factor: $y^2 - 16 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 + 2y + 1 =$

(2) Factor: $x^2 - 49 =$

(3) Factor: $y^2 - 81 =$

(4) Factor: $8y^2 + 6y =$

(5) Factor: $x^2 - 81 =$

(6) Factor: $y^2 + 3y + 2 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 25 =$

(2) Factor: $y^2 + 5y + 6 =$

(3) Factor: $12x^2 + 8x =$

(4) Factor: $x^2 + 3x + 2 =$

(5) Factor: $8y^2 + 10y =$

(6) Factor: $x^2 + 6x + 5 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 + 8y + 15 =$

(2) Factor: $x^2 + 7x + 10 =$

(3) Factor: $y^2 + 9y + 20 =$

(4) Factor: $x^2 - 64 =$

(5) Factor: $x^2 + 4x + 4 =$

(6) Factor: $y^2 - 25 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $x^2 - 36 =$

(2) Factor: $x^2 - 25 =$

(3) Factor: $12y^2 + 20y =$

(4) Factor: $y^2 + 10y + 24 =$

(5) Factor: $y^2 + 4y + 4 =$

(6) Factor: $x^2 + 9x + 20 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $y^2 + 6y + 5 =$

(2) Factor: $32x^2 + 24x =$

(3) Factor: $12x^2 + 16x =$

(4) Factor: $x^2 - 25 =$

(5) Factor: $y^2 + 6y + 5 =$

(6) Factor: $x^2 - 81 =$

Section 4. Factorisation I

Factor each expression completely.

(1) Factor: $15y^2 + 9y =$

(2) Factor: $y^2 - 36 =$

(3) Factor: $x^2 + 9x + 18 =$

(4) Factor: $x^2 - 4 =$

(5) Factor: $x^2 + 7x + 12 =$

(6) Factor: $28y^2 + 24y =$

Answer Key

JPMethod Polynomials

F31a

- (1) $(y + 5)(y - 5)$
- (2) $(y + 4)(y + 2)$
- (3) $3x(8x + 2)$
- (4) $4y(2y + 4)$
- (5) $(x + 6)(x - 6)$
- (6) $(x + 4)(x - 4)$

F32a

- (1) $3x(4x + 2)$
- (2) $(y + 4)(y - 4)$
- (3) $(x + 2)(x + 4)$
- (4) $(x + 1)(x + 5)$
- (5) $3y(3y + 3)$
- (6) $(x + 3)(x + 5)$

F31b

- (1) $(x + 3)(x - 3)$
- (2) $2x(3x + 5)$
- (3) $(y + 9)(y - 9)$
- (4) $(x + 1)(x + 1)$
- (5) $(y + 9)(y - 9)$
- (6) $(y + 4)(y + 5)$

F32b

- (1) $(x + 6)(x - 6)$
- (2) $4y(3y + 4)$
- (3) $(x + 2)(x + 5)$
- (4) $(x + 2)(x + 6)$
- (5) $(y + 1)(y + 1)$
- (6) $3x(7x + 6)$

Answer Key

JPMethod Polynomials

F35a

- (1) $(x + 4)(x - 4)$
- (2) $4y(4y + 4)$
- (3) $(x + 5)(x - 5)$
- (4) $(y + 1)(y + 1)$
- (5) $3x(7x + 4)$
- (6) $(y + 2)(y + 5)$

F36a

- (1) $(y + 5)(y - 5)$
- (2) $(x + 4)(x + 2)$
- (3) $(y + 2)(y + 6)$
- (4) $2x(6x + 2)$
- (5) $2y(4y + 5)$
- (6) $(x + 8)(x - 8)$

F35b

- (1) $(x + 3)(x + 5)$
- (2) $(x + 8)(x - 8)$
- (3) $(y + 3)(y + 5)$
- (4) $(y + 3)(y + 4)$
- (5) $3y(4y + 2)$
- (6) $(y + 1)(y + 5)$

F36b

- (1) $(y + 1)(y + 6)$
- (2) $4x(6x + 4)$
- (3) $(y + 7)(y - 7)$
- (4) $4y(6y + 4)$
- (5) $(y + 9)(y - 9)$
- (6) $3x(4x + 6)$

F37a

- (1) $4y(8y + 4)$
- (2) $3y(4y + 5)$
- (3) $(y + 3)(y + 1)$
- (4) $(y + 4)(y + 3)$
- (5) $4y(2y + 4)$
- (6) $4y(5y + 4)$

F38a

- (1) $3y(7y + 5)$
- (2) $(x + 3)(x + 3)$
- (3) $(x + 3)(x - 3)$
- (4) $(x + 1)(x + 2)$
- (5) $(x + 2)(x - 2)$
- (6) $(y + 4)(y - 4)$

F37b

- (1) $(x + 5)(x - 5)$
- (2) $(y + 3)(y + 2)$
- (3) $4x(3x + 2)$
- (4) $(x + 2)(x + 1)$
- (5) $2y(4y + 5)$
- (6) $(x + 1)(x + 5)$

F38b

- (1) $(y + 1)(y + 1)$
- (2) $(x + 7)(x - 7)$
- (3) $(y + 9)(y - 9)$
- (4) $2y(4y + 3)$
- (5) $(x + 9)(x - 9)$
- (6) $(y + 2)(y + 1)$

F33a

- (1) $(x + 4)(x + 5)$
- (2) $(x + 4)(x + 6)$
- (3) $(y + 3)(y + 4)$
- (4) $2y(4y + 5)$
- (5) $(x + 2)(x + 4)$
- (6) $(y + 9)(y - 9)$

F34a

- (1) $(x + 4)(x - 4)$
- (2) $(y + 5)(y - 5)$
- (3) $3y(6y + 5)$
- (4) $(y + 2)(y - 2)$
- (5) $2y(6y + 5)$
- (6) $3y(2y + 2)$

F33b

- (1) $3y(6y + 3)$
- (2) $(y + 3)(y + 4)$
- (3) $2y(6y + 6)$
- (4) $(x + 9)(x - 9)$
- (5) $2y(5y + 3)$
- (6) $4x(4x + 5)$

F34b

- (1) $(y + 6)(y - 6)$
- (2) $(x + 1)(x + 4)$
- (3) $3y(7y + 4)$
- (4) $(x + 8)(x - 8)$
- (5) $3x(4x + 6)$
- (6) $4y(6y + 5)$

F39a

- (1) $(y + 3)(y + 5)$
- (2) $(x + 2)(x + 5)$
- (3) $(y + 4)(y + 5)$
- (4) $(x + 8)(x - 8)$
- (5) $(x + 2)(x + 2)$
- (6) $(y + 5)(y - 5)$

F40a

- (1) $(x + 6)(x - 6)$
- (2) $(x + 5)(x - 5)$
- (3) $4y(3y + 5)$
- (4) $(y + 4)(y + 6)$
- (5) $(y + 2)(y + 2)$
- (6) $(x + 4)(x + 5)$

F39b

- (1) $3y(5y + 3)$
- (2) $(y + 6)(y - 6)$
- (3) $(x + 3)(x + 6)$
- (4) $(x + 2)(x - 2)$
- (5) $(x + 4)(x + 3)$
- (6) $4y(7y + 6)$

F40b

- (1) $(y + 1)(y + 5)$
- (2) $4x(8x + 6)$
- (3) $4x(3x + 4)$
- (4) $(x + 5)(x - 5)$
- (5) $(y + 1)(y + 5)$
- (6) $(x + 9)(x - 9)$

